

Test 19: Dirac cone v_F ? 0.125

Verdict: PASS | Generated: 2026-09-02 15:29:36 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT-geometry: zero modes at $(\pi/2, \pi/2)$ require $L/4 \gg \lambda$; gap scaling fit $R \gg 1$.

METHOD: EXACT-geometry: zero modes at $(\pi/2, \pi/2)$ require $L/4 \gg \lambda$; gap scaling fit $R \gg 1$.

Test 19 SERIES: $\lambda \in \{0.4, 0.5, 0.6\} \times L \in \{20, 40, 60, 80\}$, $\lambda = 0.5$ $R = 1.0000$, $E_{\min} \propto L^{-0.53/12.55/0.53}$ ($0.4/0.5/0.6$) ? Dirac 4? vs gapless ?0.

Result [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 19 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[15:24:25.783] [+5781.065s] build_ab_cloud_hamiltonian: 20x20 (N=400), $\lambda = 0.4000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:24:27.667] [+5782.950s] build_ab_cloud_hamiltonian: 40x40 (N=1600), $\lambda = 0.4000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:24:30.652] [+5783.934s] build_ab_cloud_hamiltonian: 60x60 (N=3600), $\lambda = 0.4000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:24:46.991] [+5802.273s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\lambda = 0.4000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:26:09.619] [+5884.901s] build_ab_cloud_hamiltonian: 20x20 (N=400), $\lambda = 0.5000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:26:11.221] [+5886.504s] build_ab_cloud_hamiltonian: 40x40 (N=1600), $\lambda = 0.5000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:26:14.202] [+5889.485s] build_ab_cloud_hamiltonian: 60x60 (N=3600), $\lambda = 0.5000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:26:30.482] [+5905.764s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\lambda = 0.5000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:27:52.829] [+5988.111s] build_ab_cloud_hamiltonian: 20x20 (N=400), $\lambda = 0.6000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:27:54.428] [+5989.710s] build_ab_cloud_hamiltonian: 40x40 (N=1600), $\lambda = 0.6000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:27:57.396] [+5992.678s] build_ab_cloud_hamiltonian: 60x60 (N=3600), $\lambda = 0.6000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

[15:28:13.935] [+6009.217s] build_ab_cloud_hamiltonian: 80x80 (N=6400), $\lambda = 0.6000$, $t = 1.00$, $W = 0.00$, torus, 0 vortices, model=:monumental

CONSOLE CAPTURE

??

TEST 19 (pass 3, SERIES): Dirac cone ? -selective 1/L series

??

4-point mod-20 grid L=20, 40, 60, 80; ? ? {0.4, 0.5, 0.6} ? all EXACT on the torus

? RUNTIME: 12 value-only solves (largest 80?) ? 2?3 min

?=0.4 L=20: E_min = 0.086901, E_min•L = 1.7380, zero_modes = 30

?=0.4 L=40: E_min = 0.025377, E_min•L = 1.0151, zero_modes = 70

?=0.4 L=60: E_min = 0.011605, E_min•L = 0.6963, zero_modes = 110

?=0.4 L=80: E_min = 0.006593, E_min•L = 0.5275, zero_modes = 150

?=0.5 L=20: E_min = 0.618034, E_min•L = 12.3607, zero_modes = 4

?=0.5 L=40: E_min = 0.312869, E_min•L = 12.5148, zero_modes = 4

?=0.5 L=60: E_min = 0.209057, E_min•L = 12.5434, zero_modes = 4

?=0.5 L=80: E_min = 0.156918, E_min•L = 12.5535, zero_modes = 4

?=0.6 L=20: E_min = 0.086901, E_min•L = 1.7380, zero_modes = 30

?=0.6 L=40: E_min = 0.025377, E_min•L = 1.0151, zero_modes = 70

?=0.6 L=60: E_min = 0.011605, E_min•L = 0.6963, zero_modes = 110

?=0.6 L=80: E_min = 0.006593, E_min•L = 0.5275, zero_modes = 150

?=0.5 fit: E_min = 0.0043 + 12.2836/L, R? = 0.999975 (v_F = b/2? = 1.9550)

E_min•L @L=80: ?=0.4 ? 0.5275 | ?=0.5 ? 12.5535 (4? = 12.566) | ?=0.6 ? 0.5275

??

?=0.5 fit R? > 0.95 (4 pts) PASS ? E_min = 0.0043 + 12.2836/L, v_F = b/2? = 1.9550

?=0.5 scale ? [8, 18] PASS ? E_min•L(L=80) = 12.553, target 4? = 12.566

?-selectivity > 2× PASS ? E_min•L(0.5) = 12.553 vs 0.4: 0.527 / 0.6: 0.527 (q=5 odd ? gapless at E=0)

Test 19c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS